

```
x=0:0.2:5.2;  
y=exp(x);  
L=trapz(x,y)
```

```
L =  
180.8727
```

```
M=quad( 'exp(x)', 0, 5.2)
```

```
M =  
180.2722
```

```
syms x  
N=int(exp(x), x, 0, 5.2)
```

```
N =  $e^{26/5} - 1$ 
```

```
double(N)
```

```
ans =  
180.2722
```

```
x=-1:6;  
y=1./(1+x.^2);  
L=trapz(x,y)
```

```
L =  
2.1608
```

```
M=quad( '1./(1+x.^2)', -1, 6)
```

```
M =  
2.1910
```

```
syms x  
N=int(1./(1+x.^2), -1, 6)
```

```
N =  
 $\frac{\pi}{4} + \text{atan}(6)$ 
```

```
double(N)
```

```
ans =  
2.1910
```

```
x = 0:0.2:1;  
y = x .* sin(pi*x);  
L = trapz(x,y)
```

```
L =  
0.3078
```

```
M = quad( 'x.*sin(pi*x)', 0, 1)
```

```
M =  
0.3183
```

```
syms x  
N = int(x*sin(pi*x),x,0,1)
```

```
N =
```

$$\frac{1}{\pi}$$

```
double(N)
```

```
ans =  
0.3183
```

```
x = 0:0.25:2;  
y = exp(2*x);  
L = trapz(x,y)
```

```
L =  
27.3551
```

```
M = quad('exp(2*x)',0,2)
```

```
M =  
26.7991
```

```
syms x  
N = int(exp(2*x),x,0,2)
```

```
N =
```

$$\frac{e^4}{2} - \frac{1}{2}$$

```
double(N)
```

```
ans =  
26.7991
```